

INT422 IT Project Management

Unit II

Resource Scheduling & Resource Leveling

Lecture 13

Outline

- Resource Scheduling
- Resource scheduling in 5 steps
- Resource Leveling
- When to use resource leveling
- Resource leveling examples

What is resource scheduling, and why is it important?

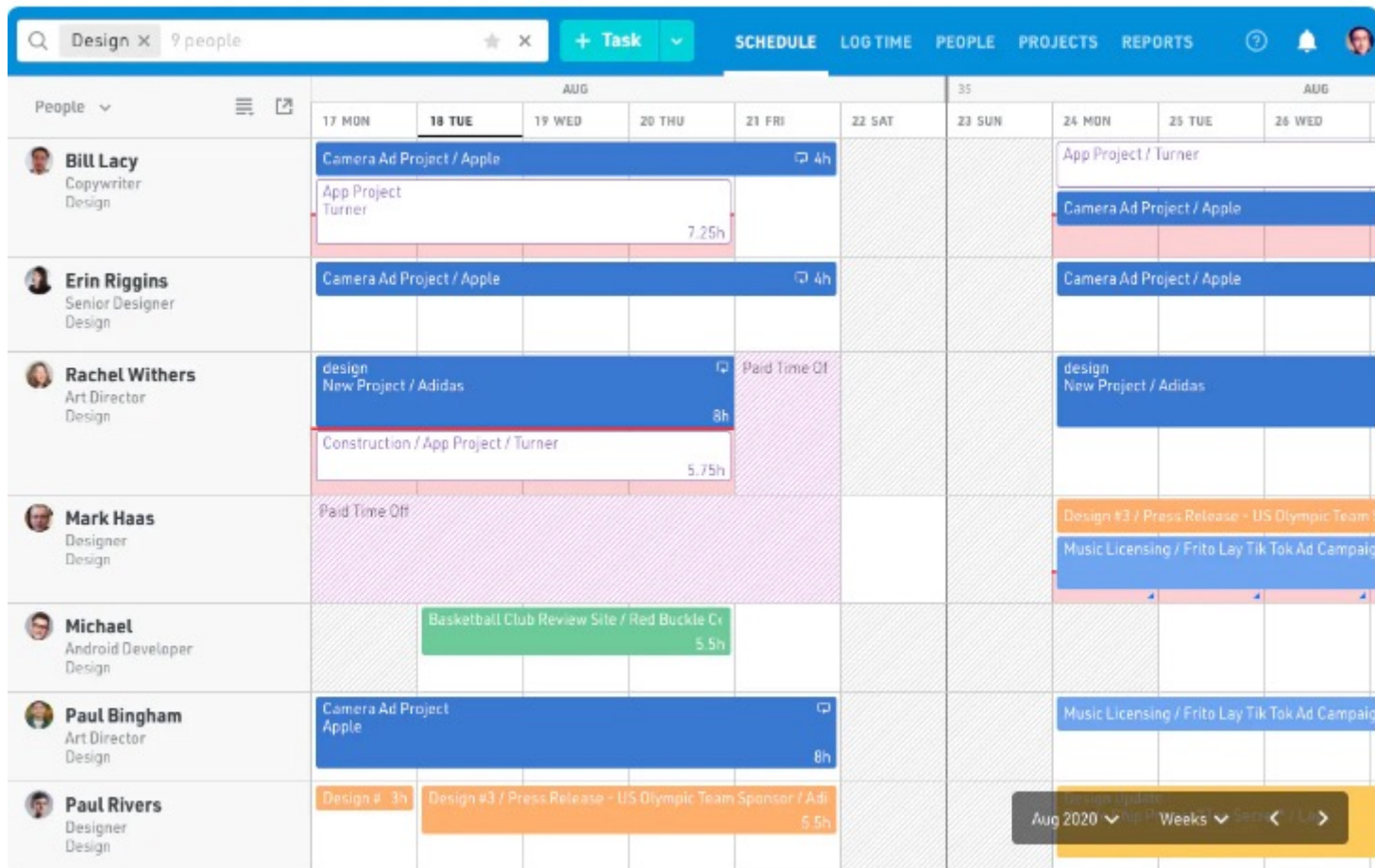
- Resource scheduling is a process used by teams to organize and structure their employees so the tasks they need to complete are scheduled based on availability and capability.
- In return, team members are always working on an optimized schedule (without being pushed to their limits), and project managers have greater flexibility should something go awry.

Resource scheduling can be used to:

-  Assign and monitor projects from start to finish
-  Schedule people based on skills and availability
-  Assess time delays and speed bumps so you can reshuffle tasks and deadlines to stop a project from going off the rails
-  Analyze how well resources are being utilized and reassign tasks to people who are not working to their full utilization rates
-  Track project estimations and outcomes to make future scheduling easier

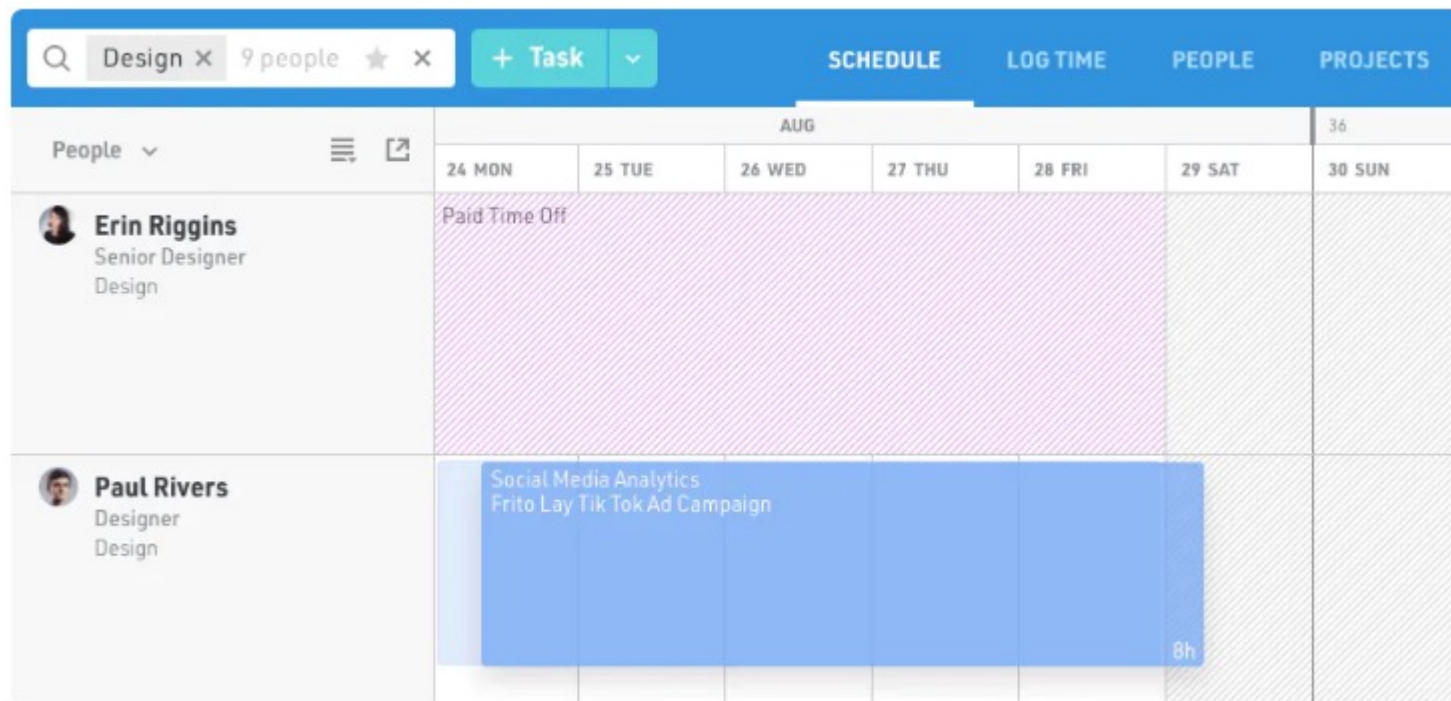
Resource scheduling in 5 steps

- Step 1. Break down tasks within a project
- Step 2. Look at your resource capabilities



Resource scheduling in 5 steps

- Step 3. Schedule tasks to team members based on their availability
- Step 4. Manage and monitor resources as the project progresses



- Step 5. Track the actual time spent on tasks and activities

What is resource leveling?

- Resource leveling is a project management technique that involves resolving over allocation or scheduling conflicts to ensure a project can be completed with the available resources.
- Resources include the time, materials, or tools needed to complete a project.



When to use resource leveling

- Resource leveling prevents resources from being spread too thin while ensuring projects still have what they need to succeed.
- This technique contributes to not only project management, but also helps maintain work-life balance for the team.
- Here are some reasons why your team can benefit from resource leveling.
 - To optimize your resources
 - To minimize deficits
 - To prevent task overloading
 - To ensure the quality of a project output

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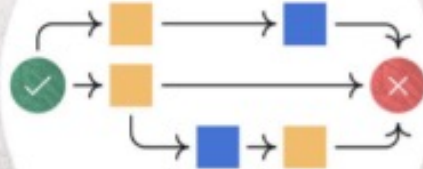
What is resource leveling?

- Depending on your team needs, here are possible outcomes:
- If the goal is to keep the current project deadline, more resources may need to be made available.
- If the goal is to run the project with currently available resources, the deadline of the project may be extended.

Project resource leveling

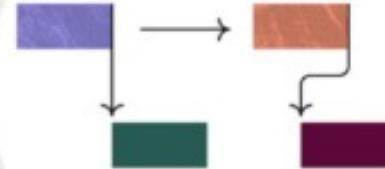
Resource leveling strategies

Critical path method



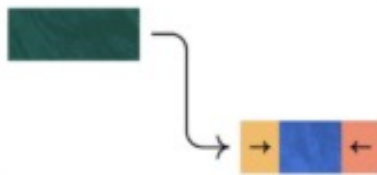
Calculates project duration and scheduling flexibility based on critical and non-critical activities

Critical chain method



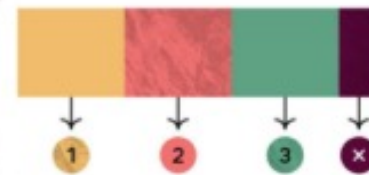
Calculates project duration based on task dependencies and resource limitations

Crashing



Shortens project timelines by assigning additional resources

Fast tracking



Speeds up projects by running activities in parallel

Resource leveling examples

Example: Delaying a project start date

- A project requires a designer to create mock-ups for the client, but the designer was double-booked, and the rest of the design team is occupied as well.
- They decide to start the project two days later when the designer is available.
- Since the designer works fast, the project end date remains the same.